

TIC 42011712

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 1.296543 d
Epoch (t0) 1683.8329 BTJD
Duration 0.1200 d
Depth 1.253 % (12,533 ppm)
SNR 80.6

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

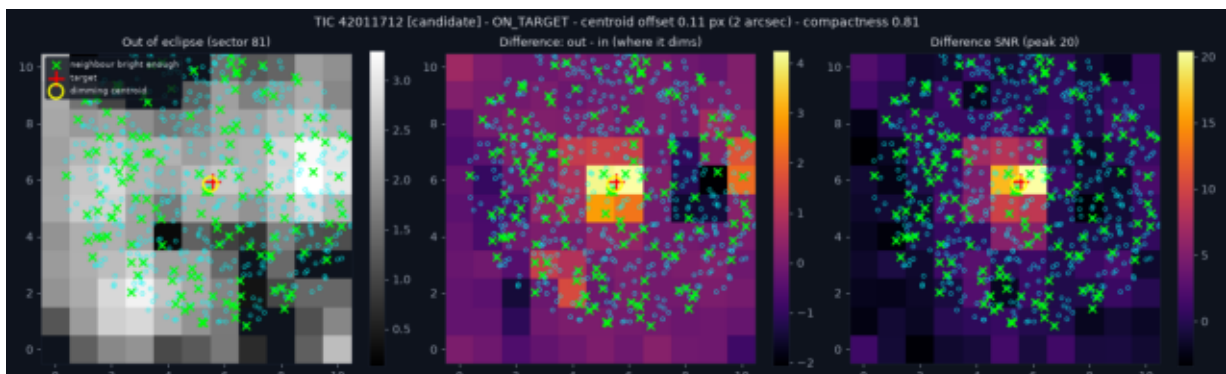
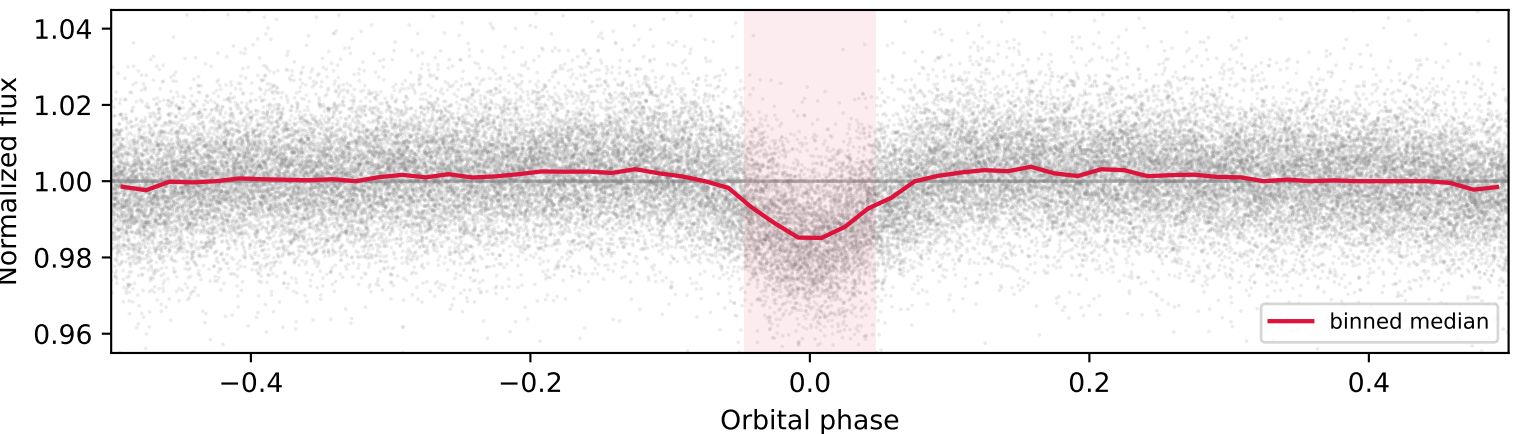
Eclipses observed 157 (gate ≥ 3)
Robust depth ratio 0.97 (gate ≥ 0.50)
Robust depth sigma 64.9 (gate ≥ 7.0)
Odd/even z-score 0.39 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 81 light-curve dip in this sector: 30.7 sigma t0 refined by 0.0100 phase
Difference-image peak SNR 20.49 (gate ≥ 5.0) Compactness 0.81 (gate ≥ 0.50)
Dimming centroid offset 0.11 px (2 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 1970538519
0.27 px away, dTmag +3.61 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.